

Lecture 1:-
Basics
JavaScript:-

HTML, CSS, JavaScript

HTML:- Website structure
CSS:- Styling
JavaScript:- functionality

ECMAScript :- official name of JavaScript.

JavaScript:- A technology that we use to create websites. It is how we use amazon, YouTube on our computer. If we were using a phone we use an app.

JavaScript was created by Brendan Eich in 1995 at Netscape communications.

JavaScript original name was Mocha, then renamed to LiveScript

After that the name JavaScript was adapted because of the popularity of Java at the time.

order of operation/ Order of precedence

precedence	operator	description
1)	()	grouping
2)	++, --	increment and decrement
3)	**	Exponentiation
4)	*,/,%	multiplication, divide, mod
5)	+, -	addition and subtraction

When operators have the same precedence, javascript evaluates them from left to right

lecture 2:-

How to link a JavaScript file to your HTML file

We will use src attribute in our script tag and give the path to the file we want to link.

```
<body>  
  <script src = "/index.js"></script>  
</body>
```

Data types:-

There are two type of data types

- 1) Primitive
- 2) Non primitive

1) Primitive:- Primitive types in javascript are the most basic data types

Primitive data types are stored in Stack.

They store single, simple values. They are immutable(their value cannot be changed directly). They are copied by value and not by reference.

there are 7 types of primitive data types

1) Number:- Represents both integers and floating point numbers

```
let age = 25;  
let price = 99.99;
```

2) String:-

Represents text

```
let name = "vaishvik";  
let city = 'Shimla';  
let msg = `Hello ${name}`;
```

3) Boolean:-

Represents logical values:-

```
let isLoggedIn = true;  
let isAdmin = false;
```

4) Undefined:-

A variable declared but not assigned a value.

```
let x;
```

5) Null:-

Represents intentional absence of value

```
let selectedUser = null;
```

6) BigInt

used for very large integers beyond the safe limit of Number

```
let bigNumber = 34235234235234324235234;
```

7) Symbol:-

used to create unique identifiers

```
let id = Symbol("userId");
```

```
// = assignment operator
// == comparison operator(doesn't check the data type and only the value)
// === strict comparison operator(checks the data type first and then the value)
```

2) Non primitive data types:-

Stored in heap memory

Non primitive data types are also called reference types

They store collections of values.

They are mutable(can be changed after creation)

Types of Non Primitive data types:-

1) Object:-

A collection of key-value pairs.

```
let person1 = {
  name : "vaishvik",
  city : "Shimla",
  rating : {
    stars : 4.5,
    count : 400
  }
};
```

* Is mutable

```
let obj2 = obj1; // pass by reference
obj2.name = "Shree";
console.log(obj1.name);
```

* Is stored by reference

2) Arrays:-

Arrays are special types of objects

3) Functions:-

Functions are also objects in JavaScript

```
function greet(name){
  console.log(`hello ${name}`);
};
```

```
greet("vaishvik");
```

4) Date:-

```
let today = new Date();
```

5) RegExp:-

```
let pattern = /abc/;
```

6) Map and set(ES6)

```
let map = new Map();
```

```
let set = new Set();
```

